

MATH 123 LECTURE NOTES

WEEK X

IMPROPER INTEGRALS

$$I = \int_a^b f(x) dx$$

① We may have $a = -\infty$ or $b = \infty$ or both

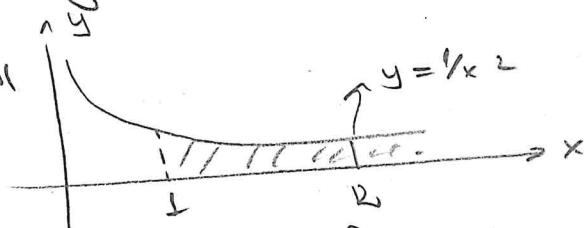
② f may be unbounded as x approaches a or b or both

Integrals satisfying ① are called improper integrals of type ①

Integrals satisfying ② are called improper integrals of type ②

Ex Find the area of the region A lying under the curve $y = 1/x^2$ and above x -axis to the right of $x=1$

$$A = \int_1^{\infty} \frac{dx}{x^2} \rightarrow \text{improper integral type ①}$$

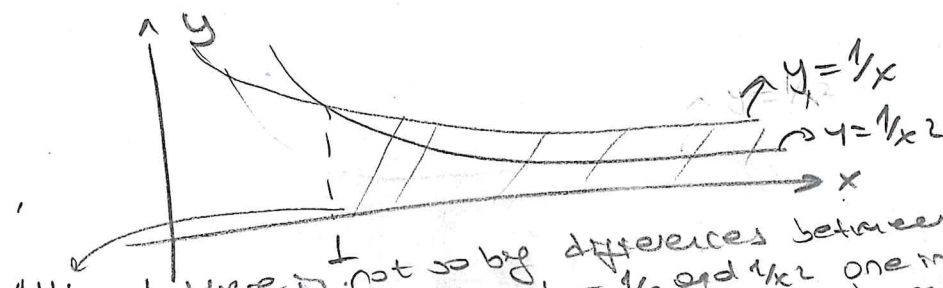


$$A = \lim_{R \rightarrow \infty} \int_1^R \frac{dx}{x^2} = \lim_{R \rightarrow \infty} \int_1^R x^{-2} dx = \lim_{R \rightarrow \infty} \left[-\frac{1}{x} \right]_1^R$$

$$= \lim_{R \rightarrow \infty} \left(-\frac{1}{R} + 1 \right) = 1$$

Ex Find the area of the region under $y = 1/x$, above $y=0$ and to the right of $x=1$

$$y = 1/x \wedge y = 1/x^2$$



$$\frac{1}{x} = \frac{1}{x^2}$$

$$x^2 = x$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$x = 0 \quad x = 1$$

Although there are not so big differences between areas under $1/x$ and $1/x^2$ one integral diverges to ∞ the other.

$$A = \int_1^{\infty} \frac{dx}{x} = \lim_{R \rightarrow \infty} \int_1^R \frac{dx}{x} = \lim_{R \rightarrow \infty} \left[\ln x \right]_1^R$$

$$\textcircled{1} = \lim_{R \rightarrow \infty} \ln R = \infty$$

IMPROPER INTEGRALS TYPE (I)

→ If f is continuous on $[a, \infty)$, we define the improper integral of f over $[a, \infty)$ as a limit of proper integrals.

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

Similarly, if f is continuous on $(-\infty, b]$, then we define

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

In both cases, if the limit exists (is a finite number), we say that the improper integral converges; if the limit doesn't exist, we say that the improper integral diverges.

If the limit is ∞ (or $-\infty$), we say that the improper integral diverges to infinity (or diverges to negative infinity).

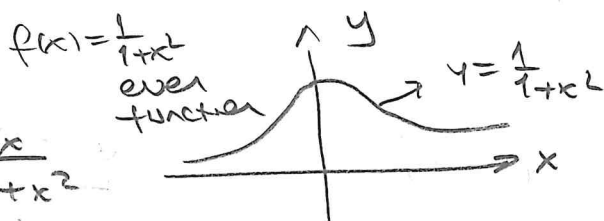
→ The integral $\int_{-\infty}^{\infty} f(x) dx$ is for f continuous on the real line, improper of type I at both endpoints. We break it into two separate integrals:

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx$$

The integral on the left converges if and only if both integrals on the right converge

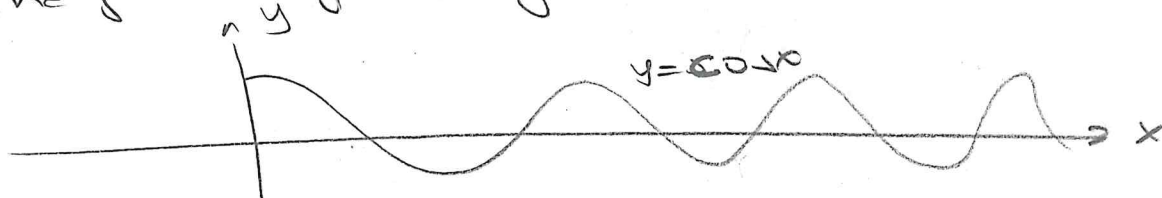
Ex $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = ?$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx &= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx \\ &= 2 \lim_{b \rightarrow \infty} \int_0^b \frac{dx}{1+x^2} = 2 \lim_{b \rightarrow \infty} \arctan b = 2 \cdot \frac{\pi}{2} = \pi \end{aligned}$$



Ex 7. $\int_0^{\infty} \cos x dx = \lim_{R \rightarrow \infty} \int_0^R \cos x dx = \lim_{R \rightarrow \infty} \sin R$ Limit DOES NOT EXIST.

The given integral diverges.



IMPROPER INTEGRALS TYPE (II)

- If f is continuous on the interval $(a, b]$ and is possibly unbounded near a , we define the improper integral

$$\int_a^b f(x) dx = \lim_{c \rightarrow a^+} \int_c^b f(x) dx$$

- Similarly, if f is continuous on $[a, b)$ and is possibly unbounded near b we define

$$\int_a^b f(x) dx = \lim_{c \rightarrow b^-} \int_a^c f(x) dx$$

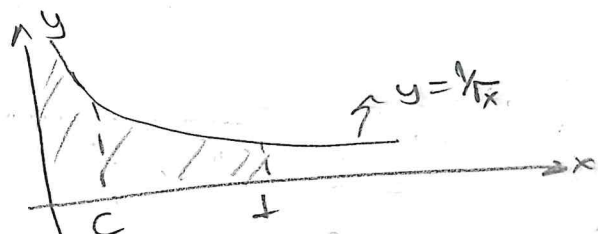
- f is discontinuous at c where $a < c < b$ and continuous on $[a, c) \cup (c, b]$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

These integrals may converge (is finite), diverge to infinity, or diverge to minus infinity.

- Ex 7 Find the area of the region S lying under $y = \frac{1}{\sqrt{x}}$ above the x -axis between $x=0$ and $x=1$.

$$A = \int_0^1 \frac{1}{\sqrt{x}} dx \rightarrow \text{Type II.}$$



$$A = \lim_{c \rightarrow 0^+} \int_c^1 x^{-1/2} dx$$

$$= \lim_{c \rightarrow 0^+} 2 \cdot x^{1/2} \Big|_c^1$$

$$= \lim_{c \rightarrow 0^+} (2 - 2\sqrt{c}) = 2$$

(3)

**
NOTE Be alert on the singularities

$$\int_{-1}^1 \frac{\ln|x|}{\sqrt{1-x}} dx = \int_{-1}^0 \frac{\ln|x|}{\sqrt{1-x}} dx + \int_0^{1/2} \frac{\ln|x|}{\sqrt{1-x}} dx + \int_{1/2}^1 \frac{\ln|x|}{\sqrt{1-x}} dx$$

$\ln 0$ is undefined $\ln 0$ is undefined $\sqrt{1-1} = 0$

Each integral on the right is improper because of a singularity at one endpoint

Ex 7 Evaluate following integrals

$$\begin{aligned} \textcircled{a} \quad \int_0^1 \frac{1}{x} dx &= \lim_{c \rightarrow 0^+} \int_c^1 \frac{1}{x} dx = \lim_{c \rightarrow 0^+} \ln|x| \Big|_c^1 \\ &= \lim_{c \rightarrow 0^+} \left(\ln \frac{1}{1} - \ln|c| \right) = \infty \end{aligned}$$

$\lim_{c \rightarrow 0^+} \ln|c| = -\infty$

This integral diverges to infinity

$$\begin{aligned} \textcircled{b} \quad \int_0^2 \frac{1}{\sqrt{2x-x^2}} dx &= \int_0^2 \frac{1}{\sqrt{1-(x-1)^2}} dx & u=x-1 \\ & & du=dx \\ &= \int_{-1}^1 \frac{1}{\sqrt{1-u^2}} du \\ &= 2 \int_0^1 \frac{1}{\sqrt{1-u^2}} du & \text{(by symmetry of even function)} \\ &= 2 \lim_{c \rightarrow 1^-} \int_0^c \frac{1}{\sqrt{1-u^2}} du \\ &= 2 \lim_{c \rightarrow 1^-} \arcsin u \Big|_0^c = 2 \lim_{c \rightarrow 1^-} \arcsin c \\ &= \pi \end{aligned}$$

$$\begin{aligned}
 \textcircled{c} \quad \int_0^1 \ln x \, dx &= \lim_{c \rightarrow 0^+} \int_c^1 \ln x \, dx \\
 &= \lim_{c \rightarrow 0^+} (x \ln x - x) \Big|_c^1 = \lim_{c \rightarrow 0^+} (0 - 1 - c \ln c + c) \\
 &= -1 + 0 - \lim_{c \rightarrow 0^+} \underbrace{\frac{\ln c}{1/c}}_{\text{L.H}} \left(\frac{-\infty}{\infty} \right) \\
 &= -1 - \lim_{c \rightarrow 0^+} \frac{1/c}{-1/c^2} \\
 &= -1 - \lim_{c \rightarrow 0^+} (-c) = -1 + 0 = -1
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{d} \quad \int_0^3 \frac{dx}{(x-1)^{2/3}} &= \int_0^1 \frac{dx}{(x-1)^{2/3}} + \int_1^3 \frac{dx}{(x-1)^{2/3}} \\
 &\quad \underbrace{x=1 \text{ undefined}} \\
 &= \lim_{b \rightarrow 1^-} \int_0^b \frac{dx}{(x-1)^{2/3}} + \lim_{c \rightarrow 1^+} \int_1^c \frac{dx}{(x-1)^{2/3}} \\
 &= \lim_{b \rightarrow 1^-} 3(x-1)^{1/3} \Big|_0^b + \lim_{c \rightarrow 1^+} 3(x-1)^{1/3} \Big|_1^c \\
 &= \lim_{b \rightarrow 1^-} (3(b-1)^{1/3} + 3) + \lim_{c \rightarrow 1^+} (3[(c-1)^{1/3} - 0]) \\
 &= 3\sqrt[3]{2}
 \end{aligned}$$

THEOREM p-integrals

If $0 < a < \infty$, then

$$\textcircled{a} \quad \int_a^\infty x^{-p} \, dx = \begin{cases} \text{converges } \frac{1-p}{p-1} & \text{if } p > 1 \\ \text{diverges to } \infty & \text{if } p \leq 1 \end{cases}$$

$$\textcircled{b} \quad \int_0^a x^{-p} \, dx = \begin{cases} \text{converges } \frac{1-p}{1-p} & \text{if } p < 1 \\ \text{diverges to } \infty & \text{if } p \geq 1 \end{cases}$$

Thm A comparison theorem for integrals.

Let $-\infty \leq a < b \leq \infty$, and suppose that functions f and g are continuous on the interval (a, b) and satisfy $0 \leq f(x) \leq g(x)$. If $\int_a^b g(x) dx$ converges, then so does

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

Equivalently, if $\int_a^b f(x) dx$ diverges to ∞ , then

so does $\int_a^b g(x) dx$

Ex Show that $\int_1^{\infty} e^{-x^2} dx$ converges and find an upper bound for its value.

On $[1, \infty)$ $x^2 \geq x$ so $-x^2 \leq -x$ and

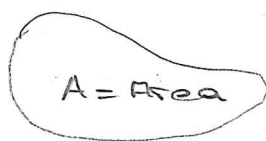
$0 < e^{-x^2} \leq e^{-x}$. Thus,

$$\begin{aligned} 0 < \int_1^{\infty} e^{-x^2} dx &\leq \int_1^{\infty} e^{-x} dx = \lim_{R \rightarrow \infty} \left. \frac{e^{-x}}{-1} \right|_1^R \\ &= \lim_{R \rightarrow \infty} \left(\frac{1}{e} - \frac{1}{e^R} \right) = \frac{1}{e} \end{aligned}$$

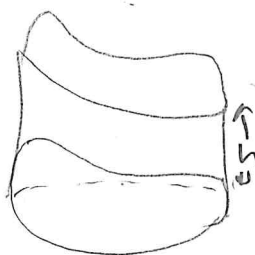
Hence, $\int_1^{\infty} e^{-x^2} dx$ converges and its value is not greater than $1 + (1/e)$.

APPLICATIONS OF INTEGRATION

VOLUMES

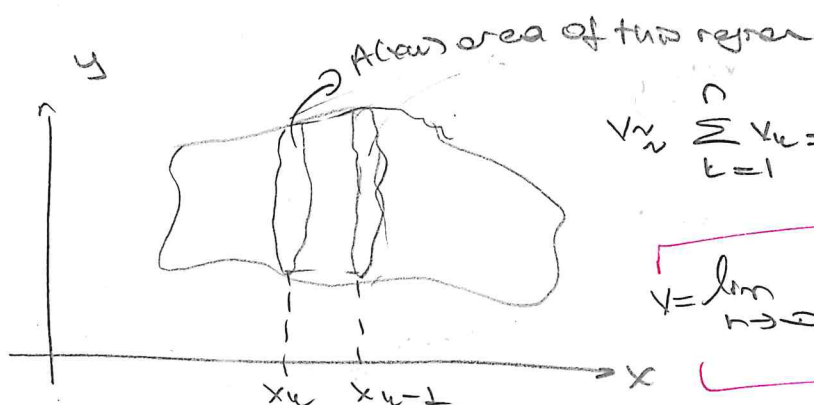


Plane whose area is known.



Cylindrical solid based on a

region
Volume = base area $\times$ height = $A \cdot h$



$$V \approx \sum_{k=1}^n V_k = \sum_{k=1}^n A(x_k) \cdot \underbrace{\Delta x_k}_{x_k - x_{k-1}}$$
$$V = \lim_{n \rightarrow \infty} \sum_{k=1}^n A(x_k) \Delta x_k = \int_a^b A(x) dx$$

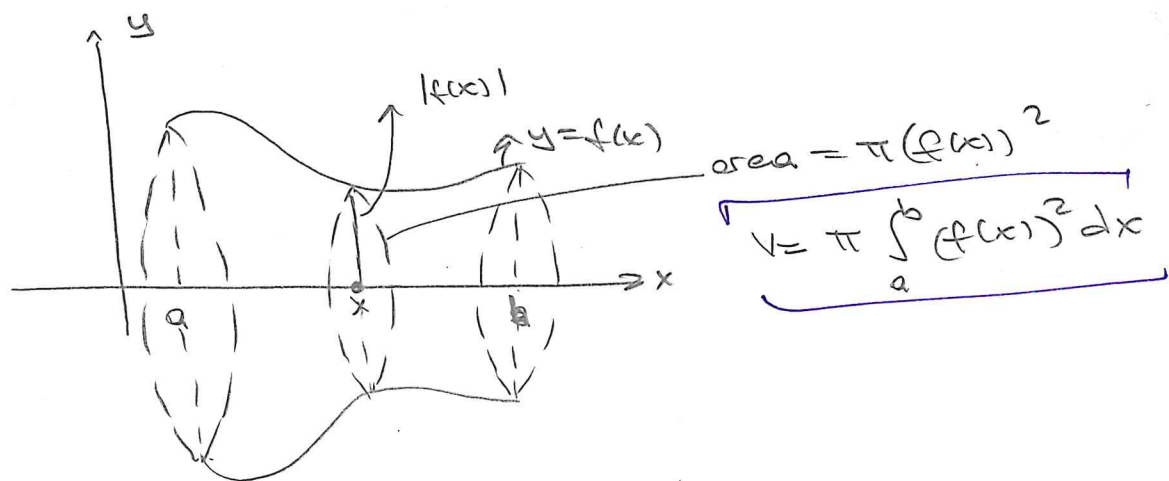
Definition The volume of a solid of integrable cross-sectional area $A(x)$ from $x=a$ to $x=b$ is the integral of A from a to b

$$V = \int_a^b A(x) dx$$

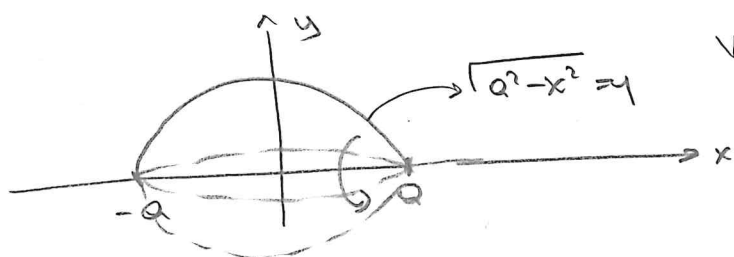
I SOLIDS OF REVOLUTION

→ If the region R bounded by $y=f(x)$, $y=0$, $x=a$ and $x=b$ is rotated about the x -axis, then the cross-section of the solid generated in the plane perpendicular to the x -axis at x is a circular disk of radius $|f(x)|$.

→ The area of this cross-section is $A(x) = \pi (f(x))^2$, so the volume of the solid revolution

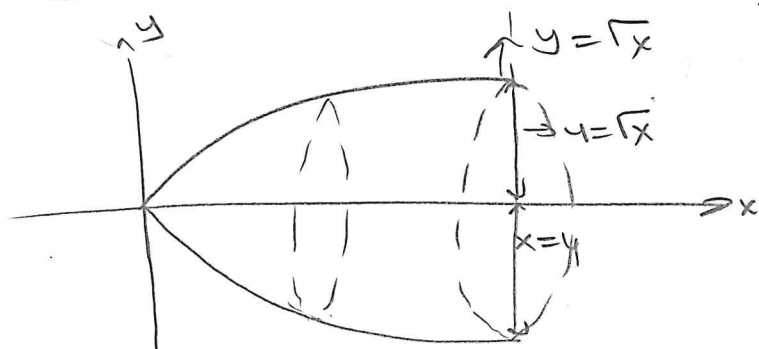


Ex 7 Find the volume of the solid ball having radius a .



$$\begin{aligned}
 V &= \pi \int_{-a}^a (\sqrt{a^2 - x^2})^2 dx \\
 &= 2\pi \int_0^a (a^2 - x^2) dx \\
 &= 2\pi \left(a^2 x - \frac{x^3}{3} \right) \Big|_0^a \\
 &= 2\pi \left(a^3 - \frac{a^3}{3} \right) = \frac{4\pi}{3} a^3
 \end{aligned}$$

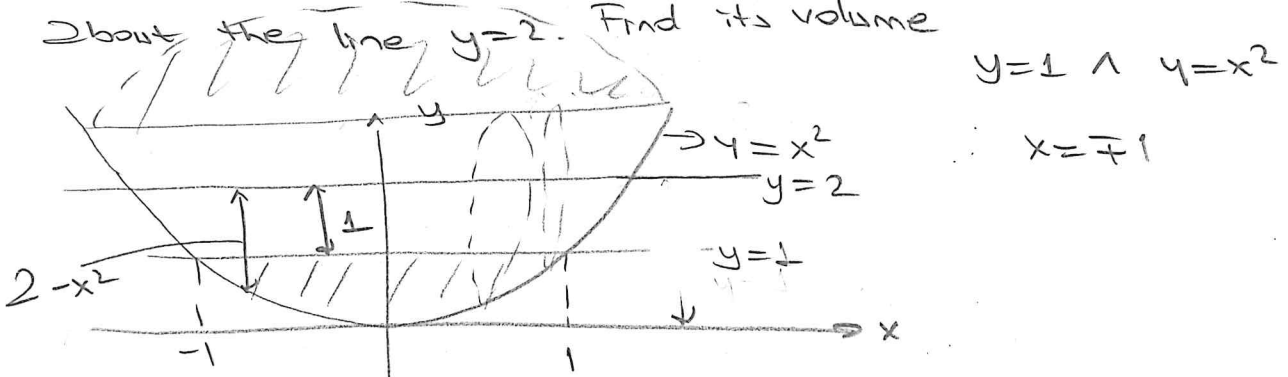
Ex 7 Find volume of the solid by rotating the region R bounded by $y = \sqrt{x}$ $0 \leq x \leq 4$ and x axis.



$$\begin{aligned}
 V &= \pi \int_0^4 (\sqrt{x})^2 dx \\
 &= \pi \int_0^4 x dx \\
 &= 8\pi
 \end{aligned}$$

Ex 7 A ring-shaped solid is generated by rotating the region R bounded by the curve $y = x^2$ and the line $y = 1$

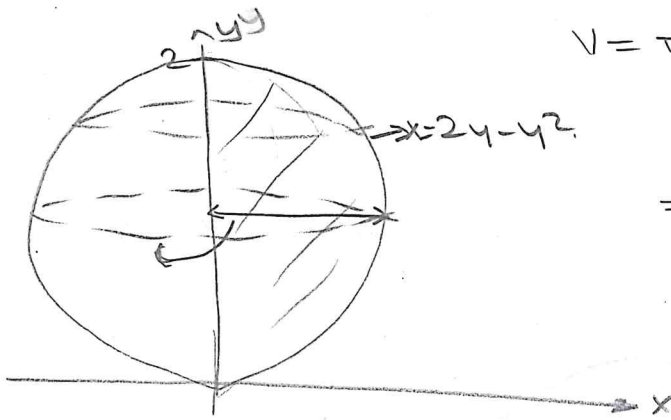
about the line $y=2$. Find its volume



$$V = \pi \int_{-1}^1 (2 - x^2)^2 - (1^2) = \pi \int_{-1}^1 (4 - 4x^2 + x^4 - 1) dx$$

$$= \pi \int_{-1}^1 (3 - 4x^2 + x^4) dx = \pi \left(3x - \frac{4x^3}{3} + \frac{x^5}{5} \right) \Big|_{-1}^1 = \frac{56\pi}{15}$$

Ex 7 Find the volume of the solid obtained by revolving the region to the right of the y -axis and to the left of the curve $x = 24 - y^2$ about y axis.



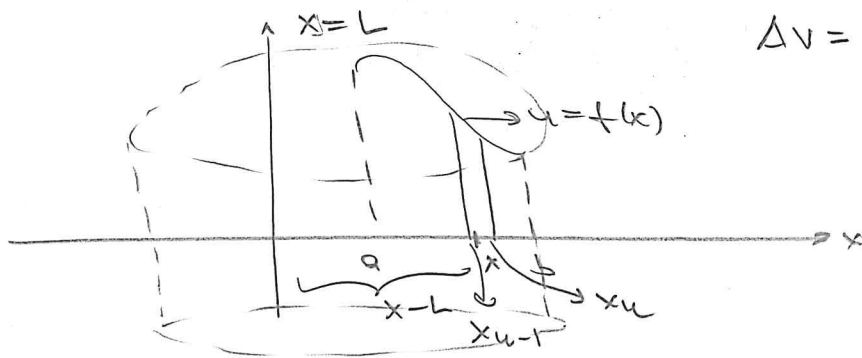
$$V = \pi \int_0^2 (24 - y^2)^2 dy$$

$$= \pi \int_0^2 (4y^2 - 4y^3 + y^4) dy$$

$$= \pi - \left(\frac{4y^3}{3} - \frac{4y^4}{4} + \frac{y^5}{5} \right) \Big|_0^2$$

$$= \frac{16\pi}{15}$$

II CYLINDRICAL SHELLS

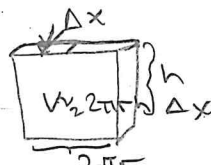


$$\Delta V = 2\pi \cdot \underbrace{\text{radius}}_{\substack{\text{Distance} \\ \text{to the} \\ \text{rotated} \\ \text{about axis} \\ (x-L)}} \cdot \underbrace{\text{height}}_{f(x)} \cdot \underbrace{\text{thickness}}_{\Delta x}$$

$$= 2\pi (x-L) f(x) \Delta x$$



$$A = 2\pi r h$$



The volume of the solid obtained by rotating the plane region $0 \leq y \leq f(x)$, $0 \leq a \leq x \leq b$ about the y -axis is

$$V = 2\pi \int_a^b x f(x) dx$$

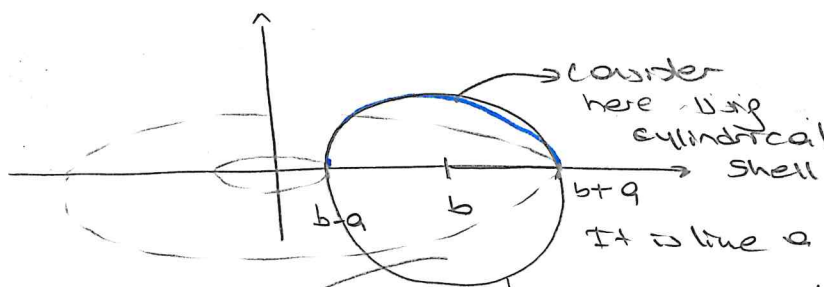
If rotating about $x=L$

The center also $L-x$ keep positivity

$$V = 2\pi \int_a^b (x-L) f(x) dx$$

Ex 7

(The volume of a torus) A disk of radius a has a center at the point $(b, 0)$ where $b > a > 0$. The disk rotated about y -axis generate a torus find its volume



$$(x-b)^2 + y^2 = a^2$$

$$V = 2 \times 2\pi \int_{b-a}^{b+a} x \sqrt{a^2 - (x-b)^2} dx$$

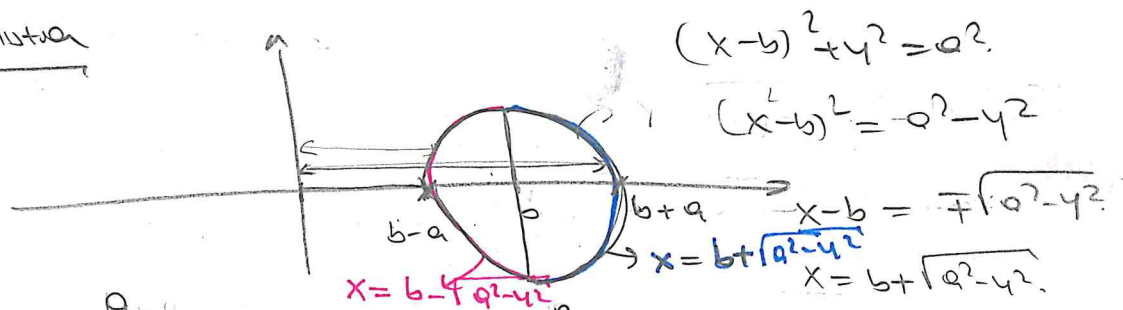
$$u = x-b$$

$$du = dx$$

$$= 4\pi \int_{-a}^a (u+b) \sqrt{a^2 - u^2} du$$

$$= 4\pi \int_{-a}^a u \sqrt{a^2 - u^2} du + 4\pi b \int_{-a}^a \sqrt{a^2 - u^2} du = 4\pi b \cdot \frac{\pi a^2}{2} = 2\pi^2 a^2 b$$

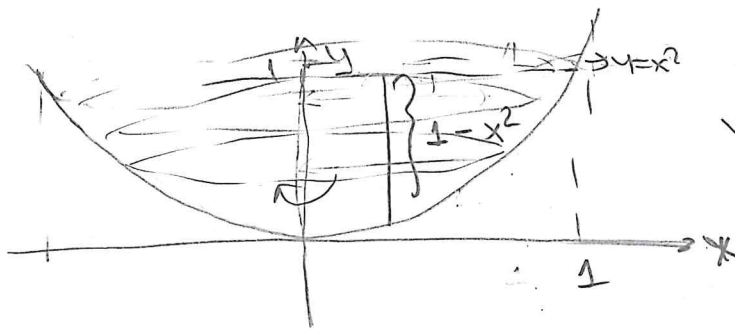
Solids of revolution



$$\begin{aligned}
 V &= \pi \left[\int_{-a}^a (b + \sqrt{a^2 - y^2})^2 dy - \int_{-a}^a (b - \sqrt{a^2 - y^2})^2 dy \right] \\
 &= \pi (2\pi a^2 b) = 2\pi^2 a^2 b.
 \end{aligned}$$

(Not in Stewart's book's (ee)notes)

Ex 7 Find volume of the bowl obtained by revolving the parabola $y = x^2$, $0 \leq x \leq 1$ about y axis



Solids of revolution

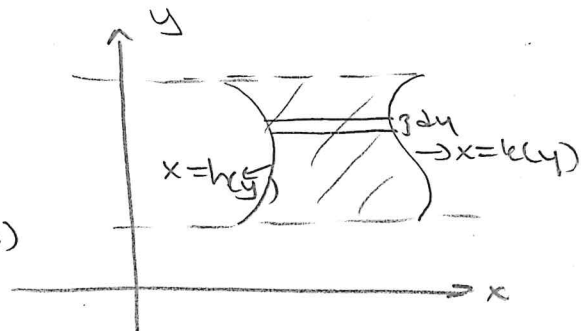
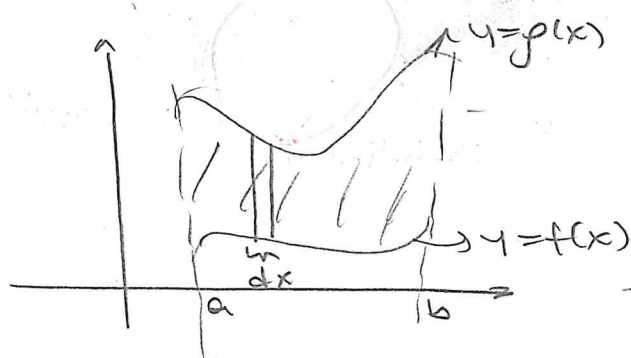
$$\begin{aligned}
 V &= \pi \int_0^1 \sqrt{y} dy \\
 &= \pi \left[\frac{y^{3/2}}{3/2} \right]_0^1 = \frac{2\pi}{3}.
 \end{aligned}$$

Cylindrical shells

$$\begin{aligned}
 V &= 2\pi \int_0^1 x (1-x^2) dx = 2\pi \left(\frac{x^2}{2} - \frac{x^4}{4} \right) \Big|_0^1 \\
 &= \pi/2.
 \end{aligned}$$

Radius
 distance
 to y axis

VOLUMES TABLE



Use Solids of Revolution

Use Cylindrical Shell

rotated
x axis
($y=0$)

$$V = \pi \int_a^b ((g(x))^2 - (f(x))^2) dx$$

$$V = 2\pi \int_c^d y (k(y) - h(y)) dy$$

rotated
y axis
($x=0$)

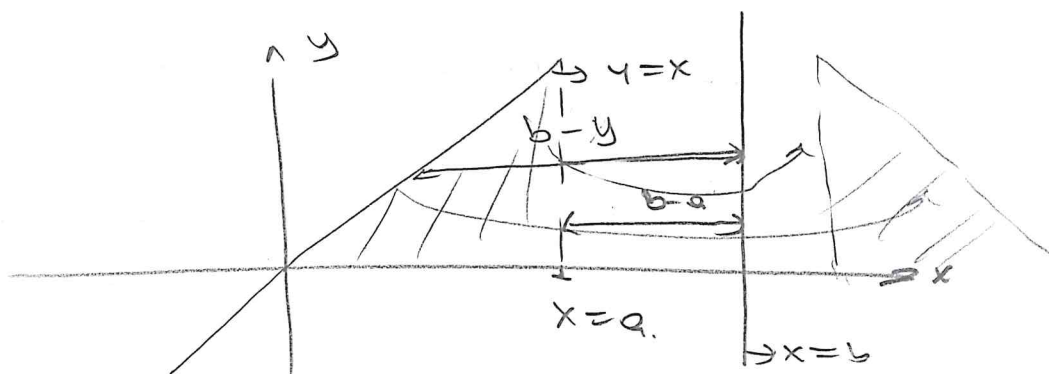
Use cylindrical shell

$$V = 2\pi \int_a^b x (g(x) - f(x)) dx$$

Solids of revolution

$$V = \pi \int_c^d ((k(y))^2 - (h(y))^2) dy$$

Ex 7 The region bounded by $y=x$, $y=0$ and $x=a>0$ rotated about $x=b>a$ line. Find volume



Cylindrical shell

$$V = 2\pi \int_0^a \underbrace{(b-x)}_{\text{radius}} x \cdot dx = 2\pi \left(\frac{bx^2}{2} - \frac{x^3}{3} \right) \Big|_0^a = \pi \left(a^2b - \frac{2a^3}{3} \right)$$

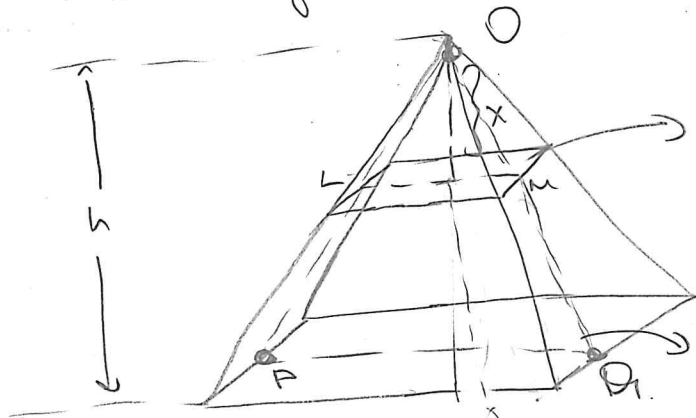
Solids of revolution

$$V = \pi \left[\int_0^a (b-y)^2 dy - \int_0^a (b-a)^2 dy \right]$$

$$= \pi \left[\left(b^2y - by^2 + \frac{y^3}{3} \right) \Big|_0^a - (b-a)^2 a \right]$$

$$= \pi \left[b^2a - \frac{1}{3}a^3 - ba^2 + 2a^3 - a^3 \right]$$

Ex 7 Verify the formula for the volume of a pyramid with rectangular base of area A and height h .



What is area $A(x)$

Use triangles OLM and OPQ

Area A $\frac{LM}{PQ} = \frac{x}{h}$

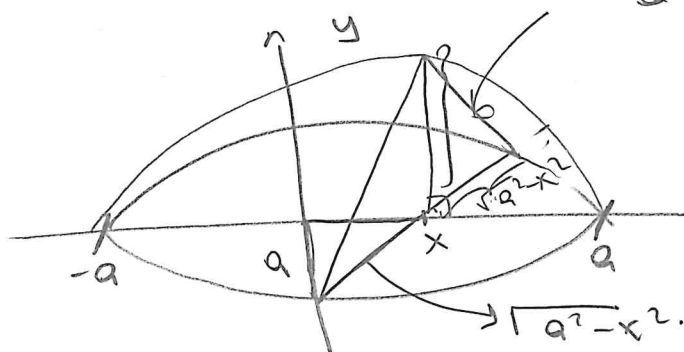
$$LM = \frac{x}{h} \cdot PQ$$

Therefore, if the area of ~~base~~ region with PQ is

$$A \quad A(x) = \left(\frac{x}{h}\right)^2 \cdot A$$

$$V = \int_0^h \left(\frac{x}{h}\right)^2 \cdot A dx = \frac{A}{h^2} \cdot \frac{x^3}{3} \Big|_0^h = \frac{1}{3} Ah.$$

Ex 7 A tent has a circular base of radius a metres and is supported by a horizontal ridge ~~bar~~ ^{str. downwards} held at height b metres above a diameter of the base by vertical supports at each end of the diameter. The material of the tent is stretched tight so that each cross section perpendicular to the ridge bar is a isosceles triangle. Find volume



We have a triangle here

$$V = \int_{-a}^a b \sqrt{a^2 - x^2} dx$$

$$= b \int_{-a}^a \sqrt{a^2 - x^2} dx$$

$$= b \frac{\pi a^2}{2} = \frac{\pi a^2 b}{2}$$

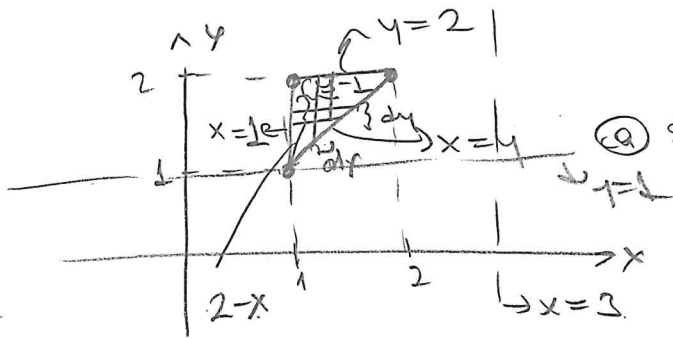
$$A(x) = \frac{1}{2} (2 \sqrt{a^2 - x^2}) b = b \sqrt{a^2 - x^2}$$

(23)

Ex (Not in BS text have's le notes)

Consider the triangle with vertices $(1,1), (1,2), (2,2)$ obtain volumes rounded around

- (a) x -axis
- (b) y -axis
- (c) $x=3$ -axis
- (d) $y=1$ -axis



(a) Solids of revolution

$$V = \pi \int_1^2 (2^2 - x^2) dx$$

Cylindrical shells rounded x -axis

$$V = 2\pi \int_1^2 y(y-1) dy$$

(b) y -axis

Solids of revolution

$$V = \pi \int_1^2 (y^2 - 1^2) dy$$

Cylindrical shells

$$V = 2\pi \int_1^2 x(2-x) dx$$

rounded y -axis

(c) $x=3$

Solids of revolution

$$V = \pi \int_1^2 ((3-1)^2 - (3-y)^2) dy$$

Cylindrical shells

$$V = 2\pi \int_1^2 \underbrace{(3-x)}_1 (2-x) dx$$

distance to rounding axis

(d) $y=1$

Solids of revolution

$$V = \pi \int_1^2 (2-1)^2 - (x-1)^2 dx$$

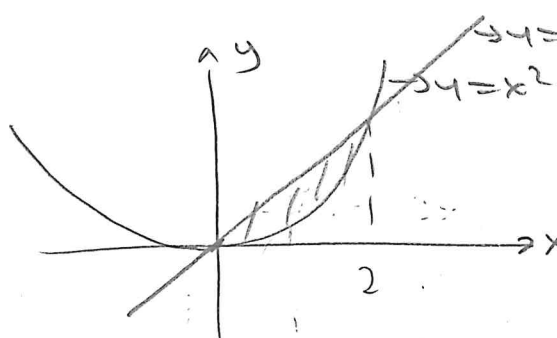
Cylindrical shells

$$V = 2\pi \int_1^2 (y-1)(y-1) dy$$

Ex (Not Bolert hoo's lec notes)

Region bounded by $y=x^2$ and $2x$ in the first quadrant.

- Ⓐ $x=0$ Ⓑ $y=2x$ Ⓒ $y=-2$ Ⓓ $y=5$ Ⓔ $x=-1$
Ⓕ $x=4$



$$2x = x^2 \quad x=0 \quad x=2$$

Revolution

Shell

Ⓐ $V = \pi \int_0^2 ((2x)^2 - (x^2)^2) dx$

$V = 2\pi \int_0^4 y \left(\sqrt{y} - \frac{y}{2} \right) dy$

Ⓑ $V = \pi \int_0^4 \left((\sqrt{y})^2 - \left(\frac{y}{2} \right)^2 \right) dy$

$V = 2\pi \int_0^2 x (2x - x^2) dx$

Ⓒ $V = \pi \int_0^2 ((2x+2)^2 - (x^2+2)^2) dx$

$V = 2\pi \int_0^4 (y+2) \left(\sqrt{y} - \frac{y}{2} \right) dy$

Ⓓ $V = \pi \int_0^2 ((5-x^2)^2 - (5-2x)^2) dx$

$V = 2\pi \int_0^4 (5-y) \left(\sqrt{y} - \frac{y}{2} \right) dy$

Ⓔ $V = \pi \int_0^4 \left((\sqrt{y}+1)^2 - \left(\frac{y}{2}+1 \right)^2 \right) dy$

$V = 2\pi \int_0^2 (x+1) (2x - x^2) dx$

Ⓕ $V = \pi \int_0^4 \left(\left(4 - \frac{y}{2} \right)^2 - \left(4 - \sqrt{y} \right)^2 \right) dy$

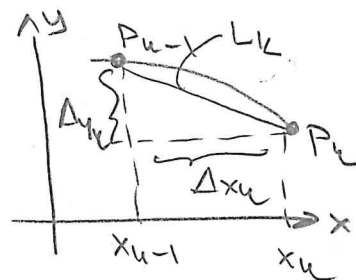
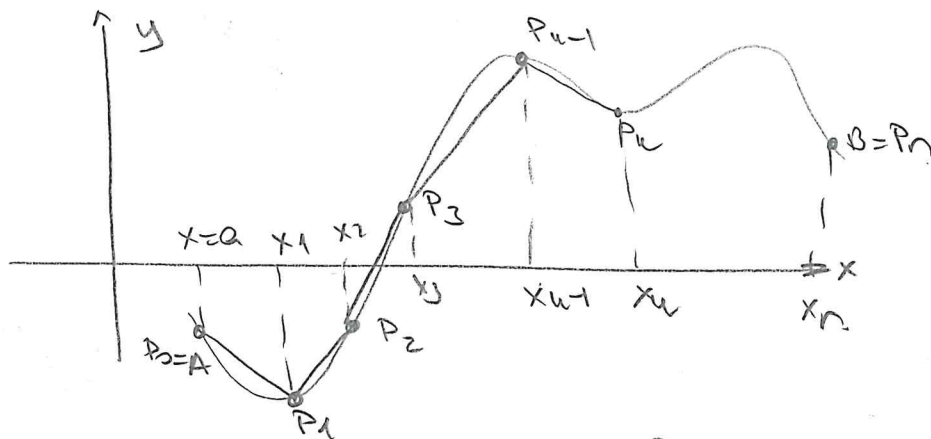
$V = 2\pi \int_0^2 (4-x) (2x - x^2) dx$

Cylindrical

Around $x=L$ your radius $(x-L)$ or $(L-x)$ and the differences between y values

Around $y=L$ your radius $(y-L)$ or $(L-y)$ and differences between x values.

ARC LENGTH AND SURFACE AREA



$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

$$\text{Arc length} = \sum_{k=1}^n L_k = \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

By mean value theorem we have a point c_k with $x_{k-1} < c_k < x_k$ such that $\Delta y_k = f'(c_k) \Delta x_k$

$$\begin{aligned} \lim_{k \rightarrow \infty} \sum_{k=1}^n L_k &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{1 + (f'(c_k))^2} \Delta x_k \\ &= \int_a^b \sqrt{1 + (f'(x))^2} dx \end{aligned}$$

Definition If f' is continuous on $[a, b]$, then the arc length of the curve $y=f(x)$ from point $A=(a, f(a))$ to $B=(b, f(b))$ is the value of the integral

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Ex Find arc length

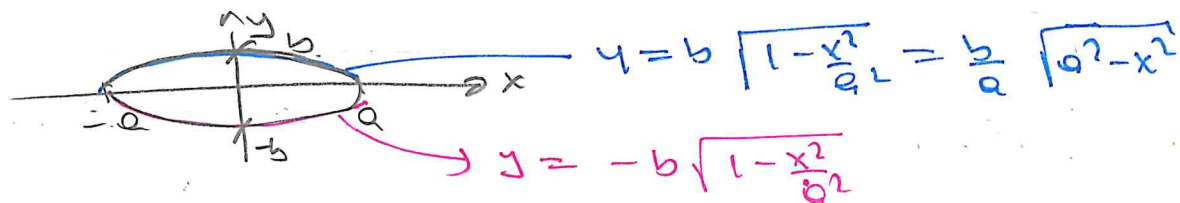
$$y = \frac{x^3}{12} + \frac{1}{x} \quad 1 \leq x \leq 4$$

$$f'(x) = \frac{x^2}{4} - \frac{1}{x^2}$$

$$\begin{aligned} 1 + (f'(x))^2 &= 1 + \left(\frac{x^2}{4} - \frac{1}{x^2}\right)^2 \\ &= 1 + \frac{x^4}{16} - \frac{x^2}{2} + \frac{1}{x^2} = \left(\frac{x^2}{4} + \frac{1}{x^2}\right)^2 \end{aligned}$$

$$L = \int_1^4 \sqrt{1 + (f'(x))^2} dx = \int_1^4 \left(\frac{x^2}{4} + \frac{1}{x^2} \right) dx = \left(\frac{x^3}{3} - \frac{1}{x} \right) \Big|_1^4 = 6$$

Ex Find length of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $a > b > 0$



$$\frac{dy}{dx} = -\frac{b}{a} \frac{x}{\sqrt{a^2 - x^2}}$$

$$1 + \left(\frac{dy}{dx} \right)^2 = 1 + \frac{b^2}{a^2} \cdot \frac{x^2}{a^2 - x^2} = \frac{a^4 - (a^2 - b^2)x^2}{a^2(a^2 - x^2)}$$

$$S = 4 \int_0^a \frac{\sqrt{a^4 - (a^2 - b^2)x^2}}{a \sqrt{a^2 - x^2}} dx$$

$$x = a \sin t \\ dx = a \cos t dt$$

$$= 4 \int_0^{\pi/2} \frac{\sqrt{a^4 - (a^2 - b^2)a^2 \sin^2 t}}{a \cdot (a \cos t)} a \cos t dt$$

$$= 4 \int_0^{\pi/2} \sqrt{a^2 - (a^2 - b^2) \sin^2 t} dt =$$

$$= 4a \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 t} dt \rightarrow \text{can't be evaluated by elementary techniques}$$

$e = \sqrt{a^2 - b^2} / a$ is the eccentricity of the ellipse

Definition Arc length of $x=g(y)$ $c \leq y \leq d$.

If g' is continuous on $[c, d]$, the length of the curve $x=g(y)$ from $A=(g(c), c)$ to $B=(g(d), d)$

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_c^d \sqrt{1 + (g'(y))^2} dy$$

Ex 7 Find length of the curve $y=(x/2)^{2/3}$ $0 \leq x \leq 2$

$$\frac{dy}{dx} = \frac{2}{3} \left(\frac{x}{2}\right)^{-1/3} \cdot \frac{1}{2} = \frac{1}{3} \left(\frac{2}{x}\right)^{1/3} \quad \text{is not defined for } x=0 \text{ can evaluate integral between } x \in [0, 2]$$

$$x=0 \Rightarrow y=0 \\ x=2 \Rightarrow y=1$$

$$\left(\frac{x}{2}\right)^{2/3} = y \Rightarrow y^{3/2} = \frac{x}{2} \quad x = 2 y^{3/2}$$

$$\begin{aligned} \frac{dx}{dy} &= 2 \left(\frac{2}{2}\right) y^{1/2} = 2 y^{1/2} \quad L = \int_0^1 \sqrt{1 + 4y} dy \\ &= \frac{1}{9} \cdot \frac{2}{3} (1+9y)^{3/2} \Big|_0^1 \\ &= \frac{2}{27} (10\sqrt{10} - 1) \end{aligned}$$

~~AREAS OF~~

The Differential Formula for Arc Length

If $y=f(x)$ and if f' is continuous on $[a, b]$, then by Fundamental Theorem of Calculus we define a new function

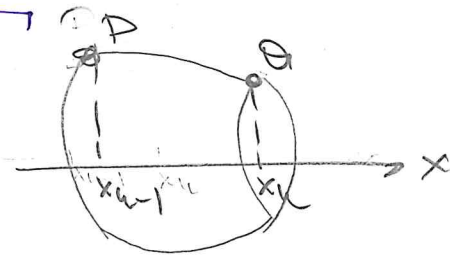
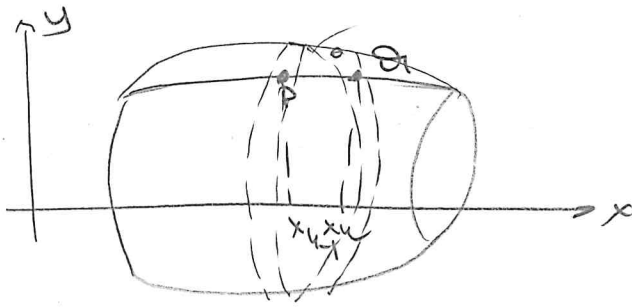
$$s(x) = \int_a^x \sqrt{1 + (f'(t))^2} dt$$

The function s is called the arc length function for $y=f(x)$

$$\frac{ds}{dx} = \sqrt{1 + (f'(x))^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \quad ds = dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$ds = \sqrt{dx^2 + dy^2}$$

AREAS OF SURFACE OF REVOLUTION



Frustum surface area
approximates

$$= 2\pi \frac{f(x_{i-1}) + f(x_i)}{2} \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$

$$= \pi (f(x_{i-1}) + f(x_i)) \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$

The surface generated by revolving the graph of a nonnegative function

$y = f(x)$ $a \leq x \leq b$ about the x -axis. The surface

is a union of the

bands like one swept out by the arc PQ .

Surface area

$$\text{Surface area} = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

Ex Find the area of the surface of a sphere of radius a .

A sphere can be generated by rotating a semicircle $y = \sqrt{a^2 - x^2}$ ($-a \leq x \leq a$) about x -axis.

$$\frac{dy}{dx} = -\frac{x}{\sqrt{a^2 - x^2}} = -\frac{x}{y}$$

$$\begin{aligned} f(x) \sqrt{1 + f'(x)^2} &= y \sqrt{1 + \frac{x^2}{y^2}} \\ &= \sqrt{y^2 + x^2} \\ &= \sqrt{a^2} \end{aligned}$$

$$S \geq 2\pi \int_{-a}^a y \sqrt{1 + \left(\frac{x}{y}\right)^2} dx$$

$$= 4\pi \int_0^a \sqrt{a^2} dx = 4\pi a x \Big|_0^a = 4\pi a^2.$$

